Module 1: Authentication & User Profiles *Handles all user access, roles, and personal records across the web and mobile platforms.*

1. users (Master Auth Table)

Column	Type	Description
user_id	UUID (PK)	Unique identifier for the account.
role	VARCHAR(20)	'Student', 'Admin', or 'Benefactor'.
username	VARCHAR(50)	Username or Student/Staff ID.
password_hash	VARCHAR(255)	Encrypted password.
email	VARCHAR(100)	Email for OTP and communications.
phone_number	VARCHAR(15)	Contact for SMS OTP/Notifications.
is_otp_verified	BOOLEAN	OTP verification status.
created_at	TIMESTAMP	Account creation date.

2. admin_profiles

Column Name	Data Type / Constraint	Description
admin_id	UUID, PK	Admin identifier
user_id	UUID, FK	Links to users
first_name	VARCHAR(50)	First name
last_name	VARCHAR(50)	Last name
department	VARCHAR(50)	e.g., OSFA
position	VARCHAR(50)	Admin role
is_archived	BOOLEAN	Soft delete flag

3. students

Column Name	Data Type / Constraint	Description
student_id	UUID, PK	Internal ID
pdm_id	VARCHAR(20), UNIQUE	Official student ID
user_id	UUID, FK	Links to users
course_id	UUID, FK	Links to academic_course
first_name	VARCHAR(50)	First name
middle_name	VARCHAR(50)	Optional
last_name	VARCHAR(50)	Last name
barangay	VARCHAR(50)	Residence
year_level	INT	Year level
gwa	DECIMAL(4,2)	Grade average
profile_photo_url	VARCHAR(255)	Photo link
is_active_scholar	BOOLEAN	Scholar flag
account_status	VARCHAR(20)	Pending/Verified/Disabled
sdo_status	VARCHAR(20)	Discipline status
is_archived	BOOLEAN	Soft delete

4. benefactors

- **benefactor_id** [UUID, PK] - Unique identifier for the benefactor.
- **user_id** [UUID, FK] - Links to users if they have a portal login.
- **organization_name** [VARCHAR(150)] - Name of the company or agency.
- **memorandum_url** [VARCHAR(255)] - Link to their MOA/agreement document.
- **is_archived** [BOOLEAN] - Defaults to FALSE. Set to TRUE to soft-delete or hide old records from active dashboards.

5. benefactor_contacts

- **contact_id** [UUID, PK] - Unique identifier.
- **benefactor_id** [UUID, FK] - Links to benefactors table.
- **contact_person** [VARCHAR(100)] - Main representative's name.

- **contact_email** [VARCHAR(100)] - Representative's email.
- **is_primary** [BOOLEAN] - supports multiple contacts per benefactor.

Module 2: System Settings & Maintenance

Global configurations and dynamic criteria managed by the admin.

6. system_ui_config *This table stores the global configurations shown in your "General" tab.*

- **setting_id** [UUID, PK] - Unique identifier (usually just one row that gets updated).
- **date_format** [VARCHAR(20)] - e.g., 'MMM DD, YYYY'.
- **timezone** [VARCHAR(50)] - e.g., 'Asia/Manila (PHT, UTC+8)'.
- **items_per_page** [INT] - Controls table pagination defaults (e.g., 10, 25, 50, 100).
- **language** [VARCHAR(20)] - e.g., 'English'.
- **ui_theme** [VARCHAR(20)] - e.g., 'Light', 'Dark', or 'System'.
- **updated_by** [UUID, FK] - Tracks the admin who last saved these changes.

7. academic_period

- **period_id** [UUID, PK] - Unique identifier.
- **current_academic_year** [VARCHAR(20)] - e.g., '2025', '2026'
- **current_semester** [VARCHAR(20)] - e.g., 'First Semester', 'Second Semester'
- **is_active** [BOOLEAN] - which period is the current one; allows history.
- **activated_by** [UUID, FK] - Links to users table.
- **activated_a**t [TIMESTAMP] - what time and date it was activated.

8. scholarship_criteria *This table holds the specific eligibility requirements for institutional programs.*

- **criteria_id** [UUID, PK] - Unique identifier.
- **scholarship_name** [VARCHAR(255)] - e.g., 'TES - Tertiary Education Subsidy'.
- **scholarship_code** [VARCHAR(20)] - e.g., 'TES', 'TDP', 'PRIVATE'.
- **min_gwa** [DECIMAL(4,2)] - The required grade to apply (e.g., 2.50, 1.75).
- **status** [VARCHAR(20)] - Shows if the criteria is 'published'.
- **is_archived** [BOOLEAN] - Defaults to FALSE. Set to TRUE to soft-delete or hide old records from active dashboards.

9. criteria_year_levels *This table is a junction table. Each eligible year level for a criteria gets its own row — fully queryable and filterable.*

- **criteria_year_level_id** [UUID, PK] - Unique identifier.
- **criteria_id** [UUID, FK] - Links to the scholarship_criteria table.
- **year_level** [INT] - year level for criteria.

10. benefactors_config *Manages external partners and their specific allocations.*

- **benefactor_config_id** [UUID, PK] - Unique identifier.
- **benefactor_id** [UUID, FK] - Links to benefactors table.
- **applicable_programs** [VARCHAR(100)] - e.g., 'BSIT / BSTM', 'All Programs'.
- **min_gwa** [DECIMAL(4,2)] - e.g., 1.75, 2.00.
- **status** [VARCHAR(20)] - 'Published' or 'Draft'.
- **is_archived** [BOOLEAN] - Defaults to FALSE. Set to TRUE to soft-delete or hide old records from active dashboards.

11. academic_course

- **course_id** [UUID, PK] - The hidden database identifier.
- **course_code** [VARCHAR(20), UNIQUE] - The short abbreviation (e.g., 'BSIT', 'BSEd').
- **course_name** [VARCHAR(150)] - The full official name (e.g., 'Bachelor of Science in Information Technology').
- **department** [VARCHAR(100)] - The college it belongs to (e.g., 'College of Computer Studies').
- **is_archived** [BOOLEAN] - Defaults to false. Set to true if the program is permanently discontinued.

Module 3: Scholarship Core System

The heart of the system tracking programs, applications, and verified scholars.

12. scholarship_programs

- **program_id** [UUID, PK] - Unique identifier for the scholarship.
- **benefactor_id** [UUID, FK] - Links to the benefactors table.
- **criteria_id** [UUID, FK] - Links to the scholarship_criteria.
- **program_name** [VARCHAR(255)] - Official name of the grant/program.
- **description** [TEXT] - Details about the scholarship.
- **allocated_slots** [INT] - Total number of available scholar slots.
- **financial_allocation** [DECIMAL(10,2)] - Budget or payout amount per scholar.
- **gwa_threshold** [DECIMAL 4,2, NULLABLE] - e.g., '2.00', '2.50'
- **requires_ro** [BOOLEAN] - TRUE if this program requires Return of Obligations.
- **renewal_cycle** [VARCHAR(20)] - 'Semester', 'Annual', or 'None'.
- **visibility_status** [VARCHAR(20)] - 'Draft' or 'Published'.
- **is_archived** [BOOLEAN] - Defaults to FALSE. Set to TRUE to soft-delete or hide old records from active dashboards.

13. program_requirements

- **requirement_id** [UUID, PK] - Unique identifier for the requirement.
- **program_id** [UUID, FK] - Links to scholarship_programs.
- **document_type** [VARCHAR(50)] - e.g., 'Grade Report', 'Enrollment Form'.
- **is_mandatory** [BOOLEAN] - TRUE if the applicant cannot proceed without it.
- **for_renewal** [BOOLEAN] - TRUE if this document is also required during the renewal phase.

14. applications (Applicants)

- **application_id** [UUID, PK] - Unique identifier for the application.
- **student_id** [UUID, FK] - Links to the students table.
- **program_id** [UUID, FK] - Links to scholarship_programs.
- **application_status** [VARCHAR(30)] - 'Pending Review', 'Interview', 'Approved', 'Rejected'.
- **evaluator_id** [UUID, FK] - Links to admin_profiles to track who reviewed it.
- **submission_date** [TIMESTAMP] - Date and time the application was finalized.

15. scholars *Tracks approved scholars and their active grant lifecycle.*

- **scholar_id** [UUID, PK] - Unique identifier for the active scholar record.
- **student_id** [UUID, FK] - Links to the students table.
- **program_id** [UUID, FK] - Links to scholarship_programs.
- **application_id** [UUID, FK] - Links to the original approved application.
- **date_awarded** [DATE] - When they officially became a scholar.

- **status** [VARCHAR(30)] - 'Active', 'Suspended', 'Graduated', 'Revoked'.
- **batch_year** [VARCHAR(10)] - e.g., '2024-2025'.

Module 4: IoT OCR & Documents

Handles all physical and digital document uploads and automated scanning.

16. documents (Polymorphic)

- **document_id** [UUID, PK] - Unique identifier for the document.
- **student_id** [UUID, FK] - Links to the student who owns the document.
- **requirement_id** [UUID, FK] - Links to program_requirements to identify the document type.
- **linked_record_id** [UUID] - Links to application_id OR renewal_id.
- **linked_record_type** [VARCHAR(30)] - 'Application' or 'Renewal'.
- **file_url** [VARCHAR(255)] - Server path to the uploaded/scanned image.
- **scanned_via_iot** [BOOLEAN] - TRUE if captured at the OSFA IoT scanner station.
- **iot_device_id** [VARCHAR(50)] - Identifier of the specific scanning kiosk used.
- **ocr_extracted_name** [VARCHAR(100)] - Raw name text extracted by EasyOCR.
- **ocr_extracted_gwa** [VARCHAR(20)] - Raw GWA text extracted by EasyOCR.
- **ocr_confidence** [DECIMAL(5,2)] - Confidence score of the OCR read (e.g., 98.50).
- **verification_status** [VARCHAR(20)] - 'Pending', 'Approved', 'Re-upload', 'Flagged'.
- **admin_notes** [TEXT] - Manual notes from the OCR Validation Hub.

17. renewals

- **renewal_id** [UUID, PK] - Unique identifier for the renewal period.
- **scholar_id** [UUID, FK] - Links to the scholars table.
- **period_id** [UUID, FK] - Links to academic_period table.
- **status** [VARCHAR(30)] - 'Pending Submission', 'Under Review', 'Approved', 'Failed'.
- **deadline_date** [DATE] - When the renewal documents must be submitted.
- **submitted_on** [TIMESTAMP] - When the student finalized their renewal upload.

Module 5: System Operations & Logistics

The day-to-day operations, including duties, payouts, discipline, and reporting.

18. return_of_obligations

- **ro_id** [UUID, PK] - Unique identifier for the obligation task.
- **scholar_id** [UUID, FK] [MODIFIED] - Links to scholars (Ensures only active scholars get ROs).
- **department_assigned** [VARCHAR(100)] - Department where the scholar will render hours.
- **task_description** [TEXT] - Details of the obligation.
- **required_hours** [INT] - Total hours to be rendered.
- **rendered_hours** [INT] - Hours already completed by the student.
- **ro_status** [VARCHAR(30)] - 'Pending', 'In Progress', 'Overdue', 'Completed'.
- **deadline_date** [DATE] - Due date for completing the obligation.

19. payout_history

- **history_id** [UUID, PK] - Unique identifier for the transaction.
- **scholar_id** [UUID, FK] [MODIFIED] - The scholar who received the money.
- **amount_received** [DECIMAL(10,2)] - The exact amount handed over or deposited.
- **date_claimed** [TIMESTAMP] - The exact moment the money was claimed.
- **processed_by** [UUID, FK] - The OSFA Admin who released the funds.
- **remarks** [TEXT] - Notes (e.g., 'Claimed via GCash', 'Late claim').

20. interviews

- **interview_id** [UUID, PK] - Unique identifier for the interview slot.
- **application_id** [UUID, FK] - Links to the specific applications.
- **interviewer_id** [UUID, FK] - Links to users (OSFA Admin/Coordinator).
- **scheduled_datetime** [TIMESTAMP] - Exact date and time of the interview.
- **location** [VARCHAR(150)] - Physical room or meeting link.
- **status** [VARCHAR(30)] - 'Scheduled', 'Completed', 'No-Show', 'Rescheduled'.
- **remarks** [TEXT] - Admin notes post-interview.

21. sdo_records

- **record_id** [UUID, PK] - Unique identifier for the disciplinary record.
- **student_id** UUID, FK] - Links to the students table.
- **reported_by** [UUID, FK] - Links to the users (SDO Staff) who filed it.
- **offense_level** [VARCHAR(20)] - 'Minor (P1)' or 'Major (P2)'.
- **offense_description** [TEXT] - Details of the violation.
- **date_reported** [TIMESTAMP] - When the offense was recorded.
- **status** [VARCHAR(20)] - 'Active' or 'Lifted'.

22. audit_logs

- **log_id** [UUID, PK] - Unique identifier for the log entry.
- **user_id** [UUID, FK] - The user who performed the action.
- **action_taken** [VARCHAR(100)] - Brief description (e.g., 'Approved Application').
- **timestamp** [TIMESTAMP] - Exact date and time of the action.
- **ip_address** [VARCHAR(45)] - Network address for security tracking.

23. generated_reports

- **report_id** [UUID, PK] - Unique identifier for the report log.
- **generated_by** [UUID, FK] - Links to admin_profiles (Who clicked generate).
- **report_type** [VARCHAR(50)] - e.g., 'CHED Scholarship Report', 'TES Consolidated Report'.
- **period_id** [UUID, FK] - Links to academic_period table.
- **file_format** [VARCHAR(10)] - 'PDF', 'Excel', or 'CSV'.
- **generated_at** [TIMESTAMP] - Exact time the report was created.
- **file_url** [VARCHAR(255)] - Server link where the generated report is stored.

24. support_tickets

- **ticket_id** [UUID, PK] - Unique identifier for the ticket.
- **student_id** [UUID, FK] - Links to the students table.
- **issue_category** [VARCHAR(50)] - e.g., 'Account Issue', 'OCR Error'.
- **description** [TEXT] - The student's explanation of the problem.
- **status** [VARCHAR(30)] - 'Open', 'In Progress', 'Resolved', 'Closed'.

- **handled_by** [UUID, FK] - Links to admin_profiles.
- **created_at** [TIMESTAMP] - When the ticket was submitted.
- **resolved_at** [TIMESTAMP] - When the admin closed the ticket.

Module 6: Communications & Mobile App

Handles internal messaging, announcements, and mobile push notification delivery.

25. messages

- **message_id** [UUID, PK] - Unique identifier for the message.
- **sender_id** [UUID, FK] - Links to the users table (Who sent it).
- **receiver_id** [UUID, FK] - Links to the users table (Who receives it, or NULL if broadcast).
- **subject** [VARCHAR(150)] - The topic of the message.
- **message_body** [TEXT] - The actual content of the message.
- **sent_at** [TIMESTAMP] - Exact date and time it was sent.
- **is_read** [BOOLEAN] - TRUE if the receiver has opened it.
- **attachment_url** [VARCHAR(255)] - Optional link if they attach a file/image.

26. announcements

- **announcement_id** [UUID, PK] - Unique identifier for the post.
- **author_id** [UUID, FK] - Links to users (OSFA staff/Benefactor).
- **subject** [VARCHAR(150)] - Headline of the announcement.
- **content** [TEXT] - Body of the announcement.
- **target_audience** [VARCHAR(50)] - 'All', 'Scholars', 'Applicants', 'TDP Only'.
- **is_ro_voluntary** [BOOLEAN] - TRUE if it includes an 'Apply' button for RO slots.
- **publish_date** [TIMESTAMP] - Scheduled or actual publish date.
- **status** [VARCHAR(20)] - 'Draft', 'Published', 'Archived'.

27. faqs

- **faq_id** [UUID, PK] - Unique identifier for the question.
- **question** [VARCHAR(255)] - The common question being asked.
- **answer** [TEXT] - The detailed answer provided by OSFA.
- **display_order** [INT] - Controls the sequence they appear on the web/app.
- **is_active** [BOOLEAN] - Allows admins to hide/show FAQs without deleting them.

28. notifications

- **notification_id** [UUID, PK] - Unique identifier.
- **user_id** [UUID, FK] [MODIFIED] - Links to users (Allows sending to admins, scholars, or benefactors).
- **type** [VARCHAR(50)] - 'Application', 'Document', 'Payout', 'RO Reminder', etc.
- **title** [VARCHAR(150)] - Short headline displayed in the push notification banner.
- **message** [TEXT] - Full message body shown inside the app.
- **reference_id** [UUID] - ID of the related record for deep linking.
- **reference_type** [VARCHAR(50)] - e.g., 'applications', 'renewals', 'documents'.
- **is_read** [BOOLEAN] - TRUE when the student opens/views the notification.
- **push_sent** [BOOLEAN] - TRUE if successfully delivered to Firebase/APNs.
- **created_at** [TIMESTAMP] - Timestamp of creation.

29. mobile_sessions

- **session_id** [UUID, PK] - Unique identifier for each mobile session.
- **user_id** [UUID, FK] - Identifies which account the session belongs to.
- **device_token** [VARCHAR(255)] - Device-specific push notification token (Firebase/APNs).
- **device_type** [VARCHAR(20)] - 'Android' or 'iOS'.
- **session_token** [VARCHAR(512)] - JWT or hashed token string used to authenticate API calls.
- **created_at** [TIMESTAMP] - Date and time the session was created.
- **expires_at** [TIMESTAMP] - Expiry timestamp of the session token.
- **is_active** [BOOLEAN] - TRUE if the session is still valid.

30. trivia

- **trivia_id** [UUID, PK] - Unique identifier.
- **question** [VARCHAR(255)] - The trivia question (e.g., "What year was PDM founded?").
- **fun_fact** [TEXT] - Additional context displayed.
- **display_date** [DATE] - The specific date this trivia should appear on the mobile dashboard.
- **is_active** [BOOLEAN] - TRUE to show in the app.

1. Return of Obligations (RO) & QR DTR System (Draft)

Table 1.1: ro_assignments *The master record connecting a scholar to a specific department for their total required hours.*

- **assignment_id** [UUID, PK] - Unique identifier.
- **scholar_id** [UUID, FK] - Links to the scholars table.
- **assigned_department** [VARCHAR(100)] - e.g., 'Library', 'Registrar', 'IT Department'.
- **department_head_id** [UUID, FK] - Links to users or admin_profiles (The person authorized to scan the QR code).
- **total_required_hours** [DECIMAL(5,2)] - e.g., 10.00 hours.
- **total_completed_hours** [DECIMAL(5,2)] - Running total (e.g., 3.50 hours).
- **semester** [VARCHAR(20)] - e.g., '1st Sem 2024'.
- **status** [VARCHAR(30)] - 'Pending Start', 'In Progress', 'Completed'.

Table 1.2: ro_dtr_logs (The QR Anti-Cheat Table) *Records every single session (even if it's just 1 hour). The scholar shows a QR code on their app, and the Department Head scans it to validate the time.*

- **log_id** [UUID, PK] - Unique identifier for the specific work session.
- **assignment_id** [UUID, FK] - Links back to ro_assignments.
- **date_rendered** [DATE] - The specific day they worked.
- **time_in** [TIMESTAMP] - When they started.
- **time_out** [TIMESTAMP] - When they finished.
- **hours_rendered** [DECIMAL(4,2)] - e.g., 1.50 (1 hour and 30 mins).
- **qr_validation_token** [VARCHAR(255)] - A unique, temporary string generated by the student's app for this session.
- **validated_by** [UUID, FK] - The specific Department Head who scanned the QR code.
- **validation_timestamp** [TIMESTAMP] - The exact second the Department Head scanned the QR code.
- **status** [VARCHAR(20)] - 'Pending Scan', 'Validated', 'Rejected'.

2. Endorsement Slip